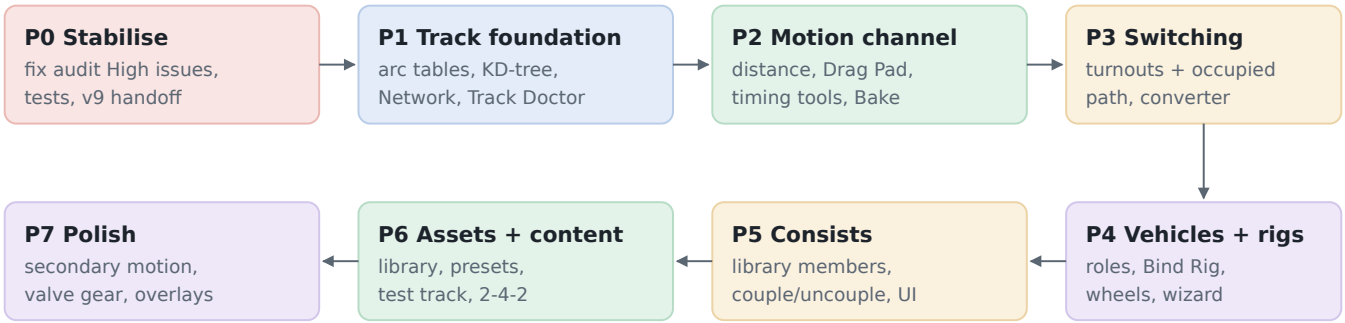
PART 11

Roadmap to completion

Every Part of this guide turned into ordered phases, each shippable and each keeping the add-on usable. It also covers the test plan that stops regressions, the decisions still open, and the first five things to do next session.

11.1 Phases at a glance



What you can do after each phase

P0: render any scene reliably; edit curves without stale offsets; two trains in one scene.
P1: build big layouts that stay fast; switches detected once; yard sidings never confused.
P2: animate a whole run with one F-curve in metres; drag with the mouse safely; farm-ready bakes.
P3: set points, shunt, switchback, back into sidings from the last car, with two trains sharing turnouts.
P4–P7: any model → vehicle in minutes; real trains from the library; the 2-4-2 with working rods.

Figure 11.1 Phase order. P4 (vehicle roles, Bind Rig) can start alongside P2/P3 if you want rigging progress sooner. It only depends on P0.

11.2 Backlog by phase

Phase	Item	Ref	Size
P0	Evaluate pivots and wagons for **all** vehicles, not only the active object	Audit #1, Part 7 §7.7	M
P0	Arc cache keyed by object and geometry hash; invalidate on edit	Audit #2, Part 4 §4.2	S
P0	Clear inherited collections on wagon clones (verify first)	Audit #4	S
P0	Pivots resolve via the existing route resolver (crossing fix)	Audit #3, Part 6 §6.5	S

Phase	Item	Ref	Size
P0	bl_info version bump, minimum Blender version, “Migrate constraints” operator	Part 1 §1.4, audit #12	S
P0	Recover the Drag Pad prototype from the fork; write the v9 handoff	Part 1 §1.4	S
P0	Headless test harness plus five baseline scenes	§11.3	M
P1	Arc tables from `interpolate_bezier`, cyclic support, KD-tree nearest	Part 4 §4.2	M
P1	Track Network + Scan Track (joints, turnouts with span test, dead ends)	Part 4 §4.3	L
P1	Track Doctor; joint auto-continue toggle; Auto Cant	Part 4 §4.5–4.8	M
P1	Split into package / Extension; UUIDs on managed objects	Part 3 §3.5, audit #8	M
P2	Consist distance as the Motion channel (a lone vehicle = consist of one)	Part 7 §7.1	L
P2	Drag Slider on distance; Drag Pad modal; step buttons; speed readout	Part 7 §7.3, 7.6	M
P2	Speed Profile, Arrive At; Bake / Clear Bake	Part 7 §7.2, 7.8	M
P3	Junction objects with keyframeable state; Junctions Ahead panel	Part 5 §5.5, 5.7	M
P3	Occupied-path evaluator with per-frame cache	Part 5 §5.5	L
P3	Convert Keyed Switches; retire keyframe mining, shadow sync, odometer log	Part 5 §5.8	M
P4	Pivot roles; chord FRAME/BODY solver; Bind/Unbind Rig	Part 6	L
P4	Wheel groups, Auto/Direct/Slip/Grip drivers; facing	Part 7 §7.4, Part 6 §6.7	M
P4	VehicleDef, Measure Vehicle, Make Vehicle wizard with templates	Part 9	L
P5	Library-based members, coupler spacing, couple/uncouple events, Train Builder UI	Part 8	L
P6	Asset library layout, test track scene, track presets, mine cart, coach, 2-4-2 + tender	Parts 4, 9, 10	L
P7	Secondary motion FX, Walschaerts helper, reverser, overlays, performance pass	Part 7 §7.5, Part 10	M

Size: S ≈ one session, M ≈ 2–3 sessions, L ≈ 4+ sessions, at the pace of the sessions so far.

11.3 Test plan

Most of the bugs found so far (the flag-ordering race, contaminated keys, stale matrix reads, the keyframe-removal iterator) were caught by hand, after the fact. A small automated suite running Blender headless catches them before you open Blender:

```
blender -b tests/scenes/switch_basic.blend -P tests/run.py -- --case all
```

Scene	Assertions
attach_roundtrip	Attach → Detach → Attach leaves the world matrix unchanged (±1e-5), with no rotation stacking.
arc_accuracy	For 20 factors, the add-on's predicted position = Blender's evaluated Follow Path position (±1 mm), including a cyclic curve and resolution 4.
switch_basic	Diverge at frame 36: positions at frames 1/35/36/37/120 match stored golden values when scrubbed forward, backward and randomly.
switchback	Main → curve 2 → reverse onto curve 3. Every pivot and wagon is on the expected curve at each checkpoint.
yard_sidings	Dead-end sidings ending close together never produce a switch (the §8.5 regression).
consist_spacing	Four wagons through two switches: the gap between consecutive members along the route stays within ±5 mm.
render_parity	With nothing selected, a background render pose at frames 50 and 120 equals the viewport pose.
edit_invalidation	Move a curve point after attaching: the next evaluation uses the new geometry.

11.4 Decisions still open

Question	Recommendation	Where
Adopt turnout state + occupied path instead of per-vehicle history?	Yes, in Phase 3, via the converter	Part 5
Per-train Route Plan overrides?	Optional, off by default	Part 5 §5.6
Live drag reads the current or the next keyed turnout state?	A setting; default “current”	Part 5 §5.6
Animate distance instead of offset_factor?	Yes	Part 7 §7.1
Vehicle root at the centre or the front coupler?	Centre, plus coupler distances	Part 6 §6.3
Turnout toe: auto-detected or a marker Empty?	Auto, with an optional override	Part 4 §4.4
Loco and tender: one vehicle or two?	Two, coupled, plus a consist preset	Part 10 §10.2
Couple/uncouple as timeline events or destructive edits?	Timeline events	Part 8 §8.4

Question	Recommendation	Where
Keep Mouse Drag at all once the Drag Pad works?	Remove it	Part 7 §7.3
Joint auto-continue: scene-wide or per vehicle?	Per junction now, per vehicle later	Part 4 §4.5

11.5 Risks

Risk	Mitigation
The modal-transform crash comes back	No handler writes during `TRANSFORM_OT_*` modals (check `window.modal_operators`); live control only through add-on-owned widgets; a test that runs a scripted grab.
Performance on large layouts	Network built once; KD-trees; per-frame route cache; profile with 50 curves and 5 trains as an acceptance target.
Blender API changes (layered Actions changed in 4.4/5.x)	Isolate F-curve access in one helper (`_track_it_find_fcurve` already does this); pin the minimum version in the manifest.
A big rewrite stalls progress	Every phase ships. The old paths stay behind toggles until the converter and tests prove the new ones.
Scenes from before the migration	Migrate/Convert operators, plus Bake as an escape hatch.

11.6 Next five actions

1. **Write the v9 handoff and bump the version**
   Record the consist prototype (not in v8), the missing Drag Pad, and this guide's audit list.

2. **Fix “active object only”**
   Loop over every object with a Track It root in both handlers. This is the smallest change with the biggest effect (renders).

3. **Fix the arc cache**
   Key it by `curve.as_pointer()` plus a geometry hash, and clear it in `depsgraph_update_post` for updated curve ids.

4. **Route pivots through `_track_it_resolve_route_position`**
   This closes v8 §3.3 with code that already exists.

5 **Build the headless test harness**

Start with attach_roundtrip, arc_accuracy and render_parity, then refactor with confidence.

CLOSING NOTE

The hard, fiddly parts are already solved: arc-length attach, crash-safe live driving, influence switching, junction-exact consist spacing. What's left is mostly **moving decisions to where they belong** (switches on the track, motion on the train, looks on the rig) and building the tools around them. The design sessions already pointed that way.